Puneeth Reddy M. Damodar

Full-Stack Software Engineer · AI-Augmented Development
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LinkedIn · GitHub · Portfolio

SUMMARY

Full-stack engineer who's shipped systems processing millions of records daily and built production LLM-powered features adopted by 5+ research institutions across 3 countries. Core contributor to Magellon, an open-source scientific data platform. Six months of daily production experience directing AI-agentic development (Claude Code) across the full stack.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, SQL

Frontend: React, Next.js, WebSockets, REST, GraphQL

Backend: FastAPI, Node.js, Spring Boot, Microservices, Plugin architectures

AI / LLMs: OpenAI & Anthropic APIs, RAG pipelines, LangChain, agentic workflows

AI-Assisted Development: Claude Code (daily driver, 6+ months), AI-augmented full-stack delivery

Infra & Data: PostgreSQL, MongoDB, Redis, AWS (EC2/S3/RDS), Docker, Jenkins CI/CD

EXPERIENCE

Senior Software Engineer · *Florida State University — Magellon Platform* Tallahassee, FL | Jan 2025 – Present

- Shifted primary development workflow to AI-agentic coding (Claude Code) — directing, reviewing, and integrating AI-generated code across the full stack while maintaining production quality and architectural ownership, significantly accelerating feature delivery.
- Built a real-time React/Next.js dashboard with WebSocket streaming for live pipeline outputs — achieved sub-second perceived latency, became the primary interface for 10+ internal teams across 3 countries.
- Redesigned PostgreSQL and MongoDB schemas (composite indexing, N+1 elimination, connection pooling) — cut query latency 30% on multi-GB datasets without downtime.
- Refactored sequential job processing into a parallel distributed model (Celery/Redis) — tripled pipeline throughput and unblocked high-volume data runs that previously had to be queued manually.
- Integrated OpenAI, Anthropic, and LangChain APIs into a RAG pipeline — built an AI assistant that reduced time-to-insight on large dataset analysis from hours to under 10 minutes.
- Coordinated feature rollouts with research teams across 3 countries, maintaining backward compatibility across 5+ active institutional deployments with zero migration-related incidents.

Software Engineer (Part-Time, concurrent with M.S.) · *Florida State University* Tallahassee, FL | Aug 2023 – Dec 2024

- Designed async data ingestion pipelines in Python with configurable processing stages, retry logic, and event queuing — replaced brittle manual import workflows that regularly failed on large datasets.
- Proposed and implemented a plugin-based backend architecture decoupling core platform logic from feature modules — let teams ship independently and reduced regression risk on new releases.
- Wrote internal developer documentation and onboarding guides that cut ramp-up time for new contributors at partner institutions.

Software Engineer · *Novartis* Remote | Aug 2021 – Dec 2022

- Built features across a centralized data management platform (React, FastAPI, Node.js, Spring Boot, PostgreSQL) consolidating 12+ siloed workflows — added RBAC, automated validation, and REST API integrations.
- Took sole ownership of a Git-style version control feature for relational data — designed the schema, built row-level diffing and branching, and delivered rollback end-to-end; became a talking point in stakeholder demos.
- Profiled and optimized API response times 35% across Node.js, Spring Boot, and FastAPI services via N+1 fixes and Redis caching — users noticed the improvement on high-traffic pages.
- Set up a 3-layer test suite (Jest, React Testing Library, Cypress) integrated into Jenkins CI/CD — cut production bug rate 75% and became the baseline for all new feature work.

Software Engineer Intern · *Novartis* Remote | Aug 2020 – Jul 2021

- Built Node.js REST APIs handling 1M+ daily records — migrated 8 legacy batch scripts to an async pipeline, cutting processing time 40% with no data loss.
- Optimized bulk DB ingestion from 8s to under 2s on 100K-row batches; shipped full-stack features across React and backend services through the team's CI/CD pipeline.

PROJECTS

LLM Input Security Browser Extension · *TypeScript, Python, FastAPI, WebSockets · 2024*

- Browser extension intercepting real-time user input across LLM platforms (ChatGPT, Claude), auto-detecting PII and credentials via an agentic LLM workflow. Full solo ownership: architecture, TypeScript/React frontend, Python/FastAPI backend.

RAG Documentation Chatbot · *Python, LangChain, FastAPI, PostgreSQL · 2024*

- Built a retrieval-augmented generation chatbot over Magellon's documentation — reduced repetitive support queries from new users and partner institutions by surfacing accurate answers from versioned docs at runtime.

EDUCATION

M.S., Computer Science · Florida State University · GPA 3.7/4.0 · Jan 2023 – Dec 2024
B.E., Computer Science · Vellore Institute of Technology · GPA 8.97/10.0 · Jul 2016 – Jun 2021